Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/23/2007
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Project Title:	Microorganisms in the Spacecraft Environment		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Environmental health		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	77204-5001	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 03/31/2007 per PI (7/07)		
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COI Name (Institution):	Willson, Richard (University of Houston)		
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	The original project had two long-term objectives: (1) To develop general and specific microbial monitoring systems that utilize technologies that can be readily implemented in spacecraft. (2) To better understand the long-term effects of the space environment on representative bacteria. Three project tasks were to address these objectives. The primary task was to develop microbial monitoring technology that can be used on long term space missions. 16S ribosomal RNA, (16S rRNA), is the single most useful molecule for bacterial identification and therefore it was the target of choice in the monitoring work. In order to develop a general sentinel system for pathogens and other problematic organisms without prior knowledge of what organism might be encountered, we sought to take advantage of our earlier discovery that there are a large number of informative subsequences in 16S rRNA (“signature sequences”) whose presence in a unknown sample provide information that can		

Task Description:	be used to provide information on phylogenetic position (genetic affinity) or specific identity. For routine monitoring we proposed to look for the presence of specific genera and species as expected indicators of anticipated problems such as contaminated water or air. In a more sophisticated mode, we would seek to determine the genetic affinity of any problematic organism without prior knowledge of what it would be. Three approaches were investigated for this purposes; (A) microarrays of 16S rRNA targeted probes; (B) mass spectrometry and (C) solution hybridization with detection by fluorescence using molecular beacons. The essence of the second task was to conduct experiments with the objective of understanding how a low gravity/high background radiation environment affects bacteria over extended time periods. The third project task which was abandoned due to the early project termination was extension of the core monitoring technology to detection of microbial escape from an advanced life support system. The task addressed multiple critical questions in the HSR&T roadmap. This included needs for methodologies to detect and monitor microbial populations relating to human disease, and biofouling of equipment. The project also sought to obtain a better understanding of how bacteria respond to and adapt to the microgravity environment of space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The mass spectrometric approach to bacterial identification that was developed is extremely fast and utilizes instrumentation that can be used for a variety of other tasks. Thus, for some applications in areas such as environmental microbiology., medical diagnosis and biodefense it may offer advantages over sequencing approaches. The low shear modeled microgravity results point out that different organisms can respond quite differently to these conditions. Similar low shear conditions are found in the human body and hence the results of studies done using the rotating bioreactor (HARV) may be more representative of how a pathogenic organism will respond to factors such as antibiotics than traditional culture studies.
Task Progress:	A major project task was to develop microbial monitoring systems that can be used in space environments. Three alternative technologies, hybridization arrays, mass spectrometry, and solution hybridization with molecular beacons were examined. Each of these methods targets information carried by 16S rRNA and will require nucleic acid purification protocols. We are, thus, also examining simplified methods for isolating and purifying nucleic acid samples that can be used in space environments. At this stage, MALDI-TOF mass spectrometry appears to be the most promising approach. The key advantage mass spectrometry is that the core instrumentation is compatible with space flight and potentially has many uses. We initially made a detailed investigation of the feasibility of using the information obtained by digesting 16S rRNA from an unknown organism with a base specific nuclease such as ribonuclease T1 and identifying the mass of all the resulting fragments. Our analysis of this scenario, (see Zhang et al., BMC- Bioinformatics, 7: 117 (2006)) identified two key issues. The first was loss of information due to the inability of mass spectrometry to distinguish fragments with the same composition but different sequence (e.g. CACAAAG vs AAAACCG). It was found that fragments of length six and larger contained useful information and that enough of such fragments would be available to allow identifications to be made. The second difficulty is that the mass of cytosine and uracil are so close together that the relatively low resolution instruments that likely would be available during a space flight would not be able to distinguish them. We therefore have implemented an alternative experimental protocol in which the rRNA is first converted to DNA and then regenerated as RNA with a mass labeled analog of uracil Using this protocol we successfully identified five out of five bacteria in test runs (Jackson et al., Intn. J. Mass Spectrom. 261: 218-226 (2007)). Most recently we began to apply the approach to mixtures. A subtraction algorithm was devised to deconvolute the spectra. This approach was applied to the analysis of the spectra obtained from a complex human cheek sample. We were able to identify the same dominant organisms that were found by a more traditional cloning and sequencing approach. In the array area, we developed an evolutionary programming approach to the design of a universal array of less than 1000 15mer signature sequence probes. No organism is uniquely identified by a single probe. Instead, the design is such that any organism will test positive with at least 40 probes. The pattern obtained then serves to determine the genetic affinity of the unknown organism (see -Chumakova et al., J. Biol. Phys & Chem., 5: 121-128 (2005)). With less than a thousand probes, this "signature chip" can in principle determine the genetic affinity of any bacterium. We have completed extensive computational validation of the concept, and developed an optimized set of signature sequences. These have been tested in small scale for methods development and optimization of spacer offset sequences, rRNA fragmentation methods, and labeling and analysis methods. Based on these results, a custom microarray was obtained from Combimatrix. Tests of this chip with E. coli resulted in a positive identification. However, many probes either failed to hybridize or gave false positive signals. This was expected because the behavior of individual hybridization probes is notoriously unpredictable from sequence information alone. These bad probes will need to be eliminated from the chip and replaced with alternatives that work. It therefore was necessary to revise the software so that it could start a new optimization using a partially specified set of probes, e.g. those that work. This is now complete and further testing of the arrays is continuing. The third approach to microbial monitoring that we expected would be useful for routine monitoring was molecular beacons. Beacons carry a fluor that is initially quenched. When reaching their target in the 16S rRNA, the beacon probe hybridizes to the 16S rRNA resulting in exposure of the fluor and generation of a strong fluorescent signal. This solution hybridization approach would be advantageous because it would require minimal sample preparation and could be conducted with easily prepackaged reagents. The key problem is that the signal produced is insufficient for identification of small numbers of bacteria. Our proposed solution to this problem was to use a “beacon swarm” in which multiple probes would target the 16S rRNA molecule in multiple signature sites but emit fluorescence at a single wavelength. With ten targets a single molecule of target would produce an order of magnitude more fluorescence than is normally produced. To date, we have not been able to successfully demonstrate feasibility of the approach; possibly because of unintended interactions between the individual beacon probes. Since molecular beacons are unlikely to resolve the routine monitoring need, we have revised efforts began during a prior NSBRI award to develop small arrays that target a modest number of informative organisms. Thus, an array is envision that would test for E. coli, enterics in general, gram positives and gram negatives in general, Staphylococcus and Streptococcus, etc. Probes were previously designed for many of these targets and tested but not in an array format. Recently we have constructed small arrays with these probes and begun testing them. Most of them work. The primary

problem is false positives. The offending probes will be removed and replaced with alternatives.

In order to perform microbial monitoring in the spacecraft environment using the methods discussed above (or alternative approaches developed by others) it will likely be necessary to have simplified procedures for obtaining and purifying nucleic acids samples. During the past year, limited work continued in two areas; the use of compaction agents to rapidly isolate RNA and DNA by selective precipitation and immobilized metal affinity chromatography (IMAC) for rapid purification.

A second major project goal was to better understand the response of bacteria to the space environment. In order to do this, we examined the growth of representative bacteria under simulated microgravity conditions using the NASA developed rotating wall bioreactor (HARV). An initial comparison of aerobic and anaerobic results revealed that there was no gene uniquely responding to these two growth conditions as would occur if the response was actually to gravity and that its response is quite different from that of its close relative Salmonella. However, reviewers raised objections to the statistical analysis employed. We therefore conducted an independent parametric analysis of the data in collaboration with Dr. Yuriy Fofanov a biostatistician in our Computer Science Department in order to address the reviewer's concerns. With this analysis completed, a paper describing our results in detail has now been published (Tucker et al., BMC Microbiology 7:15 (2007)). These studies only address the immediate bacterial response to the low shear modeled microgravity environment. We had for some time been attempting to conduct long-term growth studies in the HARV to determine if the bacterium adapts and thereby changes its response to the microgravity environment. These studies had until recently been hampered by difficulties in keeping the reactor contaminant free for extended periods. This problem has been solved and we have now grown cultures for over 1000 generations without contamination. We are hoping to analyze these cultures using other funds available to us in the near future.

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